

PARK FALLS, WI, USA

WMO: 727410

Lat: 45.955N

Lon: 90.424W

Elev: 457

StdP: 95.95

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.1	-35.2	0.1	-25.9	-30.7	0.2	-23.9	9.3	-7.2	8.4	-8.0	2.6	290	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	29.3	20.6	27.6	19.6	26.1	18.7	22.3	27.3	21.0	25.8	20.0	24.6	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	16.1	25.6	19.3	14.8	24.0	18.2	13.9	22.8	67.6	27.2	63.1	25.9	59.0	24.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.2	7.3	6.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-29.4	24.2	4.2	1.2	-32.5	25.1	-34.9	25.8	-37.3	26.5	-40.4	27.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-10.4	-8.9	-2.6	4.2	13.2	18.0	19.1	18.1	13.0	5.7	-1.9	-7.6	(6)
(7)		DBStd	11.88	6.92	7.08	6.53	5.05	5.13	4.16	3.08	3.29	4.35	4.47	5.12	6.28	(7)
(8)		HDD10.0	2836	632	528	392	185	27	2	0	0	19	148	358	544	(8)
(9)		HDD18.3	4992	890	762	648	424	172	55	28	46	166	393	607	803	(9)
(10)		CDD10.0	1037	0	0	2	11	126	241	281	251	111	13	1	0	(10)
(11)		CDD18.3	152	0	0	0	0	12	44	51	38	6	0	0	0	(11)
(12)		CDH23.3	1162	0	0	0	3	139	384	339	268	28	1	0	0	(12)
(13)		CDH26.7	243	0	0	0	0	21	101	68	52	1	0	0	0	(13)
(14)	Wind	WSAvg	2.9	3.2	3.1	3.1	3.3	2.9	2.6	2.6	2.5	2.7	2.9	3.3	3.0	(14)
(15)	Precipitation	PrecAvg	841	28	22	45	67	89	112	114	94	104	87	45	30	(15)
(16)		PrecMax	1109	67	51	91	147	158	228	232	234	211	196	111	55	(16)
(17)		PrecMin	625	10	2	8	21	53	30	34	33	26	23	7	10	(17)
(18)		PrecStd	156	14	13	24	33	30	47	51	49	54	46	26	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	6.9	18.5	23.1	28.6	31.1	31.2	30.6	25.9	20.5	15.1	4.5	(19)
(20)			MCWB	1.1	3.4	11.2	14.0	16.6	21.0	22.9	21.9	18.1	14.2	10.9	1.5	(20)
(21)		2%	DB	2.2	4.4	13.2	18.5	26.7	29.2	28.4	28.6	24.1	17.9	10.6	2.2	(21)
(22)			MCWB	0.5	1.1	7.5	10.7	16.8	19.9	21.5	21.0	17.6	11.9	8.0	0.1	(22)
(23)		5%	DB	0.4	2.2	8.9	15.6	24.7	27.4	26.7	26.5	22.3	15.7	8.1	0.9	(23)
(24)			MCWB	-0.8	-0.2	4.3	9.9	15.6	19.5	19.9	19.6	16.6	10.8	5.4	-0.4	(24)
(25)		10%	DB	-1.0	0.2	5.9	13.0	22.4	25.4	25.2	24.5	20.3	13.4	5.5	-0.3	(25)
(26)			MCWB	-2.0	-1.4	2.4	7.4	14.7	18.0	18.9	18.4	16.0	9.1	2.9	-1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	4.1	11.7	15.0	19.8	22.8	24.2	23.6	20.3	15.1	12.6	1.8	(27)
(28)			MCDWB	3.1	6.8	17.4	20.2	25.6	28.9	28.8	29.2	23.4	18.3	14.5	3.8	(28)
(29)		2%	WB	0.5	1.3	7.7	12.6	18.5	21.5	22.4	21.8	19.0	13.2	8.7	0.5	(29)
(30)			MCDWB	2.0	4.0	12.3	17.0	23.9	27.5	27.1	26.4	22.0	16.8	10.4	1.8	(30)
(31)		5%	WB	-0.7	0.0	4.8	9.9	17.3	20.2	21.1	20.5	18.0	11.2	5.4	-0.4	(31)
(32)			MCDWB	0.5	2.1	8.5	15.4	22.1	25.8	25.4	25.2	21.1	14.9	7.9	0.7	(32)
(33)		10%	WB	-2.0	-1.3	2.8	7.6	15.5	19.0	19.9	19.3	16.5	9.1	3.0	-1.3	(33)
(34)			MCDWB	-1.0	0.3	5.5	12.5	20.8	24.1	24.0	23.7	19.7	12.8	5.3	-0.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	9.9	10.3	11.3	10.3	9.2	9.9	9.5	8.9	6.6	7.3	(35)
(36)			MCDBR	9.3	10.4	14.1	14.6	14.3	12.2	11.0	12.0	11.7	13.5	10.0	6.7	(36)
(37)			MCWBR	8.3	8.3	9.2	8.8	7.2	6.4	6.0	6.2	6.7	8.8	8.1	6.0	(37)
(38)		5% WB	MCDBR	9.0	10.2	12.9	13.6	12.0	11.0	9.7	10.4	9.8	12.3	9.7	6.4	(38)
(39)			MCWBR	8.2	8.4	8.6	8.7	7.3	6.5	6.1	6.2	6.8	9.2	8.4	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.263	0.292	0.332	0.359	0.359	0.361	0.360	0.340	0.304	0.278	0.250	(40)	
(41)		taud	2.404	2.334	2.401	2.437	2.399	2.420	2.435	2.448	2.499	2.584	2.544	2.441	(41)	
(42)		Ebn at Noon	892	925	943	928	907	906	899	888	880	873	838	837	(42)	
(43)		Edh at Noon	69	91	100	106	114	113	110	104	91	72	62	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.49	2.57	3.74	4.53	5.32	5.82	6.08	5.09	3.91	2.42	1.43	1.03	(44)	
(45)		RadStd	0.16	0.23	0.34	0.47	0.40	0.43	0.25	0.34	0.24	0.28	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page